

Assignment 6

Github: https://github.com/Parth2619/Java_Lab/tree/main/Assignment%206

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Batch: AIML B1

Title: Employee Payroll System: Implement payroll calculation demonstrating single, multilevel, and hierarchical inheritance.

Code 1) EmployeePayrollSystem.java

```
import java.util.ArrayList;

abstract class Employee {
    protected int empId;
    protected String name;
    protected String PANNo;
    protected String joiningDate;
    protected String designation; // Maps to role

    public Employee(int empId, String name, String PANNo, String joiningDate, String designation) {
        this.empId = empId;
        this.name = name;
        this.PANNo = PANNo;
        this.joiningDate = joiningDate;
        this.designation = designation;
    }

    public abstract double calcCTC();

    public void printDetails() {
        System.out.println("EmpID    : " + empId);
        System.out.println("Name      : " + name);
        System.out.println("Designation : " + designation);
        System.out.println("PAN No    : " + PANNo);
        System.out.println("Joining Date: " + joiningDate);
        System.out.printf("Calculated CTC: INR %.2f%n", calcCTC());
    }
}

class FullTimeEmployee extends Employee {
    protected double baseSalary;
    protected double perfBonus;
    protected double hiringCommision;
```

```

public FullTimeEmployee(int empId, String name, String PANNo, String joiningDate, String designation,
    double baseSalary, double perfBonus, double hiringCommision) {
    super(empId, name, PANNo, joiningDate, designation);
    this.baseSalary = baseSalary;
    this.perfBonus = perfBonus;
    this.hiringCommision = hiringCommision;
}

@Override
public double calcCTC() {
    if (designation.equalsIgnoreCase("SWE")) {
        return baseSalary + perfBonus;
    } else if (designation.equalsIgnoreCase("HR")) {
        return baseSalary + hiringCommision;
    }
    // Base case for other full-time roles
    return baseSalary;
}
}

class ContractEmployee extends Employee {
    private int noOfHrs;
    private double hourlyRate;

    public ContractEmployee(int empId, String name, String PANNo, String joiningDate, String designation,
        int noOfHrs, double hourlyRate) {
        super(empId, name, PANNo, joiningDate, designation);
        this.noOfHrs = noOfHrs;
        this.hourlyRate = hourlyRate;
    }

    @Override
    public double calcCTC() {
        return noOfHrs * hourlyRate;
    }
}

class Manager extends FullTimeEmployee {
    private double TA;
    private double eduAllowance;

    public Manager(int empId, String name, String PANNo, String joiningDate, String designation,
        double baseSalary, double perfBonus, double TA, double eduAllowance) {
        // Manager extends FullTimeEmployee, hiringCommision not primarily used here
        super(empId, name, PANNo, joiningDate, designation, baseSalary, perfBonus, 0.0);
        this.TA = TA;
        this.eduAllowance = eduAllowance;
    }
}

```

```

@Override
public double calcCTC() {
    // baseSalary + perfBonus + TA + eduAllowance
    return baseSalary + perfBonus + TA + eduAllowance;
}
}

public class EmployeePayrollSystem {
    public static void main(String[] args) {
        System.out.println("=== Employee Payroll System ===");

        ArrayList<Employee> staff = new ArrayList<>();

        // SWE Example
        staff.add(new FullTimeEmployee(101, "Alice", "ABCDE1234F", "2022-01-10", "SWE",
            80000, 15000, 0));

        // HR Example
        staff.add(new FullTimeEmployee(102, "Bob", "FGHIJ5678K", "2021-05-20", "HR",
            60000, 0, 10000));

        // Contract Employee Example
        staff.add(new ContractEmployee(103, "Charlie", "KLMON9012P", "2023-03-15", "Contract SWE",
            160, 500));

        // Manager Example
        staff.add(new Manager(104, "Diana", "QRSTU3456V", "2018-11-01", "Manager",
            100000, 25000, 10000, 5000));

        // Print details and CTC for all employees
        for (Employee e : staff) {
            e.printDetails();
            System.out.println("-----");
        }
    }
}

```

Output:

=== Employee Payroll System ===

EmpID : 101

Name : Alice

Designation : SWE

PAN No : ABCDE1234F

Joining Date: 2022-01-10

Calculated CTC: INR 95000.00

EmpID : 102

Name : Bob

Designation : HR

PAN No : FGHIJ5678K

Joining Date: 2021-05-20

Calculated CTC: INR 70000.00

EmpID : 103

Name : Charlie

Designation : Contract SWE

PAN No : KLMON9012P

Joining Date: 2023-03-15

Calculated CTC: INR 80000.00

EmpID : 104

Name : Diana

Designation : Manager

PAN No : QRSTU3456V

Joining Date: 2018-11-01

Calculated CTC: INR 140000.00
